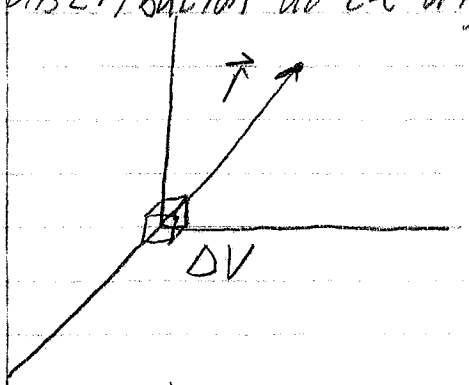
Finding retarded potentials using Fourier Transforms

Starting with the wave equation, which the scalar potential satisfies:

$$\frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} - \nabla^2 \varphi = \frac{\rho(\vec{r},t)}{\epsilon_0} \quad (1)$$

This describes the potential at a position $\vec{r}$ from a charge distribution at the origin



we replace t with the retarded time: $t \to t - \frac{|\vec{r}-\vec{r}'|}{c}$

In integral form, the solution for φ is

$$\varphi(\vec{r},t) = \frac{1}{4\pi\epsilon_0}\int dV' \frac{\rho(\vec{r}', t - \frac{|\vec{r}-\vec{r}'|}{c})}{|\vec{r}-\vec{r}'|}$$

where the prime coordinates refer to the charge distribution

In general, $\rho(\vec{r}=0,t)\Delta V = \Delta q(t)$

we shrink the charge distribution down to a point:

$\rho(\vec{r},t) = \Delta q(t)\,\delta(\vec{r})$

We now employ the Fourier transform: $\varphi(\vec{r},t) \to \varphi(\vec{k},\omega)$

$$\varphi(\vec{r},t) = \int \frac{d^3k\,d\omega}{(2\pi)^4}\,\varphi(\vec{k},\omega)\,e^{i(\vec{k}\cdot\vec{r}-\omega t)} \quad (2)$$

The right-hand side of (1) transforms thus:

$$\frac{\rho(\vec{k},\omega)}{\epsilon_0} = \int d^3\vec{r}\,dt\,e^{-i(\vec{k}\cdot\vec{r}-\omega t)}\frac{\Delta q(t)\,\delta(\vec{r})}{\epsilon_0}$$

$$= \int dt\,e^{i\omega t}\frac{\Delta q(t)}{\epsilon_0} = \frac{\Delta q(\omega)}{\epsilon_0}$$ where we let $\Delta q(\omega)$ remain as an integral

Using integration by parts, the left-hand side of (1) becomes

$$-\frac{\omega^2}{c^2}\varphi(\vec{k},\omega) + k^2\varphi(\vec{k},\omega) = \frac{\Delta q(\omega)}{\epsilon_0}$$

Solving this algebraic equation, we have

$$\varphi(\vec{k},\omega) = \frac{-1}{\epsilon_0}\frac{\Delta q(\omega)}{\omega^2/c^2 - k^2} \quad (3)$$

Now we take the inverse Fourier transform: (3 → 2)

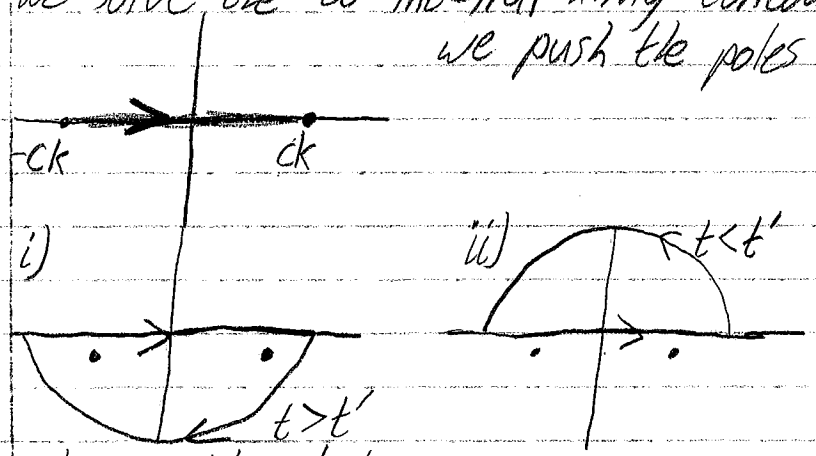
$$\varphi(\vec{r},t) = \frac{-1}{\epsilon_0}\int \frac{d^3k\,d\omega}{(2\pi)^4}\frac{\Delta q(\omega)}{\omega^2/c^2 - k^2}e^{i(\vec{k}\cdot\vec{r}-\omega t)}$$

recall that $\Delta q(\omega) = \int dt'\,\Delta q(t')e^{i\omega t'}$. Thus

$$\varphi(\vec{r},t) = \frac{-1}{\epsilon_0}\int \frac{d^3k\,d\omega\,dt'}{(2\pi)^4}\frac{\Delta q(t')}{\omega^2/c^2 - k^2}e^{i[\vec{k}\cdot\vec{r}-\omega(t-t')]}$$

We solve the ω integral using contour integration:

we push the poles off the real axis



ii) should be zero (not physical)

The ω integral becomes

$$I = \int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\frac{e^{i\omega(t-t')}}{(\frac{\omega}{c}-k+i\eta)(\frac{\omega}{c}+k+i\eta)}$$ using the Cauchy integral formula

$$= -\Sigma\, i2\pi\, \mathrm{Res}(\pm ck - i\eta)\frac{c^2}{2\pi}$$ we now let $\eta \to 0$

$$= -\frac{c^2}{2\pi}(i2\pi)\left(\frac{e^{-ick(t-t')}}{2ck} + \frac{e^{+ick(t-t')}}{-2ck}\right)$$

$$= \frac{-c}{k}\sin[ck(t-t')]$$

$$\Rightarrow \varphi(\vec{r},t) = \frac{c}{\epsilon_0}\int\frac{d^3k}{(2\pi)^3}\int_{-\infty}^{t}dt'\,\Delta q(t')\frac{\sin[ck(t-t')]}{k}e^{i\vec{k}\cdot\vec{r}}$$

Now we look at the k integral in spherical coordinates:

$$I_k = \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{k^2 \sin\theta \, d\phi \, d\theta \, dk}{(2\pi)^3} \cdot \frac{1}{k} \sin[ck(t-t')] \, e^{ikr\cos\theta}$$

$$\int_0^{2\pi} d\phi = 2\pi \int_0^\pi \sin\theta \, d\theta \, e^{ikr\cos\theta} = \int_{-1}^{1} d(\cos\theta) \, e^{ikr\cos\theta} = \frac{1}{ikr}\left(e^{ikr} - e^{-ikr}\right)$$

$$\Rightarrow I_k = \frac{1}{(2\pi)^2 r} \int_0^\infty dk \, \frac{e^{ick(t-t')} - e^{-ick(t-t')}}{2i} \cdot \frac{e^{ikr} - e^{-ikr}}{i}$$

$$= \frac{-1}{2r(2\pi)^2} \int_0^\infty dk \left[\underbrace{e^{ik[c(t-t')+r]} + e^{-ik[c(t-t')+r]}} - \underbrace{e^{ik[c(t-t')-r]} - e^{-ik[(t-t')-r]}} \right]$$

$$= \frac{-1}{2r(2\pi)^2} \left[\int_{-\infty}^{\infty} dk \, e^{ik[c(t-t')+r]} - \int_{-\infty}^{\infty} dk \, e^{ik[c(t-t')-r]} \right]$$

$$\delta(x-\alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \, e^{ik(x-\alpha)}$$

$$\Rightarrow I_k = \frac{1}{4\pi r}\left[\delta(c(t-t')-r) - \cancel{\delta(c(t-t')+r)} \right] \quad \text{not physical}$$

Thus

$$\varphi(\vec{r},t) = \frac{c}{4\pi\epsilon_0 r} \int_{-\infty}^{t} dt' \, \delta(c(t-t')-r) \, \Delta q(t')$$

$$= \frac{c}{4\pi\epsilon_0 r} \int_{-\infty}^{t} dt' \, \Delta q(t') \, \frac{1}{c} \delta\left(t' - \left(t - \frac{r}{c}\right)\right)$$

$$\Rightarrow \varphi(\vec{r},t) = \frac{\Delta q(t - \frac{r}{c})}{4\pi\epsilon_0 r}$$

The vector potential is analogous:

$$\vec{A}(\vec{r},t) = \frac{\Delta q(t - \frac{r}{c}) \vec{v}}{4\pi\epsilon_0 r}$$